

180 GBaud PAM4 Driver-Modulator Engine for IM/DD Transmissions in the O-Band

Son Tran¹, Sebastian Lauck¹, Gerrit Fiol¹, Jonathan Andree¹, Felix Ganzer¹, Robert Emmerich¹, Colja Schubert¹, Patrick Runge¹, Martin Schell^{1,2}, Ronald Freund^{1,2}, Shawn Parker³, Trong Phan³, Billy Allen³

¹Fraunhofer Heinrich-Hertz-Institute, Einsteinufer 37, 10587 Berlin, Germany

²Technical University Berlin, 10623 Berlin, Germany

³MACOM Technology Solutions, Santa Clara, CA 95050, USA

thanh.son.tran@hhi.fraunhofer.de

Abstract: A co-packaged 76 GHz InP-based Mach-Zehnder modulator and a 224 GBaud-class linear differential EML driver was demonstrated, achieving 180 GBaud PAM4 back-to-back in an IM/DD transmission for O-Band operation.

1. Introduction

The continuous increase in data traffic within short-reach and metro optical networks demands electro-optical transmitters capable of supporting high symbol rates while maintaining low power consumption and cost-effective packaging and integration [1-11]. Electro-absorption modulated lasers (EMLs) combine low energy consumption and compact footprint, making them a well-established choice for short-reach intensity modulation and direct detection (IM/DD) links [1, 6]. However, their modulation bandwidth, which is typically limited to several tens of gigahertz, together with the inherent chirp and limited extinction ratio (ER), constraint the transmission distance and degrade the signal integrity. Therefore, Mach-Zehnder modulators (MZM) have emerged as a key transmitter architecture for high-bandwidth and advanced modulation formats due to their chirp-free operation and high ER [2-5]. While MZMs on thin-film lithium niobate (TFLN) have recently demonstrated excellent bandwidth performance [7, 8], their electrode geometry typically leads to large footprint and restricts differential operation. Moreover, due to the well-known DC-drift effect [9], stable bias operation in TFLN devices requires additional permanent phase control electronics, resulting in increased power consumption. Indium phosphide (InP) platforms stand out through their ability to monolithically integrate active and passive photonic components, thus significantly reducing coupling losses, footprint and fabrication complexity [11]. Owing to its travelling-wave electrode (TWE) design, InP-based MZMs supports differential driving operation [10] and are suited for modern high-speed differential driver designs.

We introduce an InP-based MZM with a flat electro-optical response (EOR) up to 76 GHz and a modulation efficiency of $V_{\pi}L = 0.62$ V cm being co-packaged with a 20 dB gain commercial EML differential driver using a low-complexity assembly approach. The large volume availability of EML drivers allows for a cost-effective and scalable transmitter implementation. System-level measurements employing pulse amplitude modulation with 4-levels (PAM4) were performed in a back-to-back (B2B) configuration to evaluate the modulation performance in the O-band. Tab. 1 summarizes the current state-of-the-art transmitter technologies and their performance.

Reference	Technology	Component	Symbol rate/Channel	Bit rate/Channel
Lumentum (2025) [1]	InP	EML	113 GBaud PAM4	226 GBps
NTT (2025) [2]	InP	4x IQ-MZM	172 GBaud PAM6	430 GBps
Nokia (2025) [3]	InP	8x MZM	106 GBaud PAM4	212 GBps
D'Lithium (2025) [4]	SiPh	MZM+Driver	128 GBaud PAM8	384 GBps
McGill Uni (2025) [5]	TFLN	8x MZM+Driver	225 GBaud PAM4	450 GBps
This work	InP	MZM+Driver	180 GBaud PAM4	360 GBps

Tab. 1: Performance comparison of the recently presented transmitter technologies.

2. Design, Fabrication & Assembly

The p-i-n heterostructure MZM chip was fabricated on HHI's InP platform. It includes a traveling-wave electrode (TWE) design and spot size converters at the facet for simplified fiber coupling. The TWE and the on-chip termination resistors, monolithically integrated and matched to the driver's differential output impedance of 85 Ω , were optimized for differential operation with MACOM's EML driver. The radio frequency (RF) losses between the MZM and the driver chip were minimized by optimizing the interconnections to achieve the shortest possible electrical path. Therefore, both were co-packaged side-by-side on an aluminum-oxide ceramic, which not only reduced the signal

path length but also minimized the parasitic inductance introduced by the bonding wires. The ceramic submount was micro-machined to provide DC bias lines for both the MZM and the driver.

3. Measurement Setup & Results

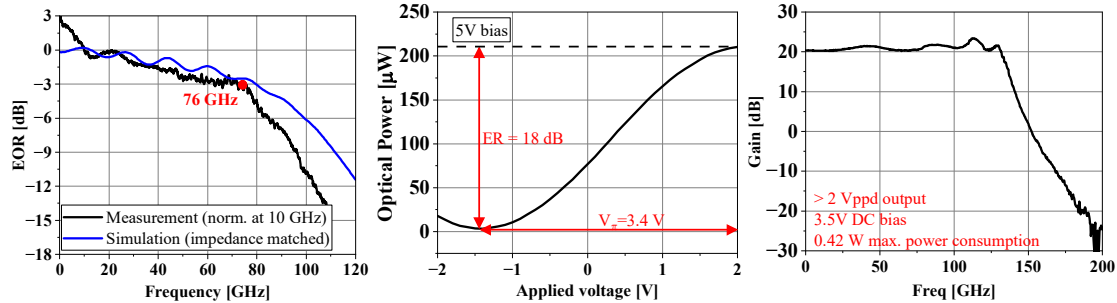


Fig. 1: Measured and simulated EO-response of MZM chip. The dip at low frequencies is due to the impedance mismatch of the 85Ω terminated MZM and the 50Ω measurement equipment (left). Measured optical transfer function of the MZM (center). Simulated S_{21} characteristic of the 20 dB linear differential driver (right).

Initially, the modulator was characterized by electro-optical (EO) chip measurements performed with a 110 GHz lightwave component analyzer. As expected, the EO-response of the MZM shows a dip at lower frequencies (Fig. 1 left), being attributed to the impedance mismatch between the 85Ω termination of the MZM and the standard 50Ω measurement instruments. The 3D- electromagnetic (EM) simulation result shows a flat response in lower frequencies with impedance of the MZM input and termination being perfectly matched. Therefore, normalizing the response beyond the dip (~ 10 GHz) is justified and yields an estimated 3 dB-bandwidth of 76 GHz and a $V_{\pi} = 3.1$ V at 5 V bias voltage. Additionally, the MZM exhibits a DC ER of 18 dB as shown in Fig. 1 (center).

The system-level measurement setup is illustrated in Fig. 2 (left). Measurements were performed at 40°C under temperature control using a thermoelectric cooler (TEC). The 20 dB gain driver (Fig. 1 right) was contacted using a 110 GHz differential RF probe in GSGSG configuration, while the DC bias lines were connected via a multi-contact probe card. The modulator subassembly was driven by a 2-channel 8-bit digital analog converter (DAC) to create a higher signal to noise ratio (SNR) for the differential drive signal. The DAC has 80 GHz of analog bandwidth and is operated at 260 GSa/s. A 1310 nm optical carrier was coupled from the fiber into the MZM. The modulated optical signal was detected by a 100 GHz photodetector (PD), being connected via a 1 mm DC-block to a 110 GHz real-time oscilloscope (RTO) for measurement. To control the received optical power at the PD, a praseodymium-doped fiber amplifier (PDFA) with optical filter and an optical attenuator were inserted between the transmitter and the PD. Furthermore, digital signal processing (DSP) with a digital root-raised cosine pulse-shaping filter with a roll-off factor of 1% was applied at both the transmitter and receiver side. Additionally, a digital feed-forward equalizer with 2048 taps was used at the receiver side to compensate for electrical and optical distortions.

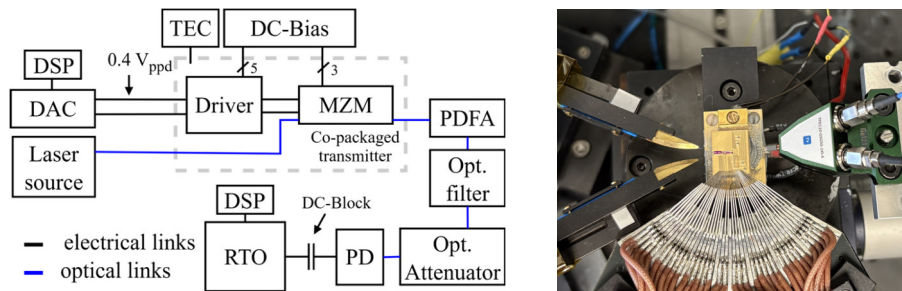


Fig. 2: Schematic of the system measurement setup (left). A picture of the driver-modulator assembly (right).

The input power of the optical carrier signal was kept constant at 15 dBm. The bit error rate (BER) was evaluated at various symbol rates as a function of the received optical power at the PD (Fig. 3 left). An optimum point at 8 dBm received optical power is observed due to saturation characteristic of the PD. Figure 3 (center and right) shows the post-equalized eye diagrams for 150 GBaud and 180 GBaud PAM4 transmission, respectively. At 150 GBaud, the eye openings remain well separated, and the BER is approximately two orders of magnitude below the 20 % overhead (OH) hard decision forward error correction (20 % HD-FEC) threshold [9]. At 180 GBaud, however, a reduction in vertical eye opening becomes evident. This degradation is primarily attributed to the limited EO-bandwidth of the

MZM (76 GHz), which constrains the optical modulation amplitude. Additional bandwidth reduction of approximately 3 dB at 80 GHz results from RF losses in the probe and the cable assembly of the measurement setup. As a result, the available electrical bandwidth of the modulator engine reached its limit. After digital equalization, residual amplitude distortion and timing jitter remain visible, indicating that the DSP has reached its compensation capability under the given operating conditions. Nevertheless, a distinguishable eye structure confirms that successful symbol detection is achievable. In summary, the co-packaged modulator engine achieved 180-GBaud PAM4 transmission with a BER below the 20 %-HD-FEC threshold under 0.4 V_{ppd} driver excitation. The total electrical energy per bit of the modulator engine is 525 mW. Note that only the power consumption of the driver, MZM and RF signal were considered. The power consumption of the DSPs, the external laser and the TEC were excluded from this estimation. At a symbol rate of 180 GBaud the electrical energy per bit of the transmitter engine is 1.45 pJ/bit.

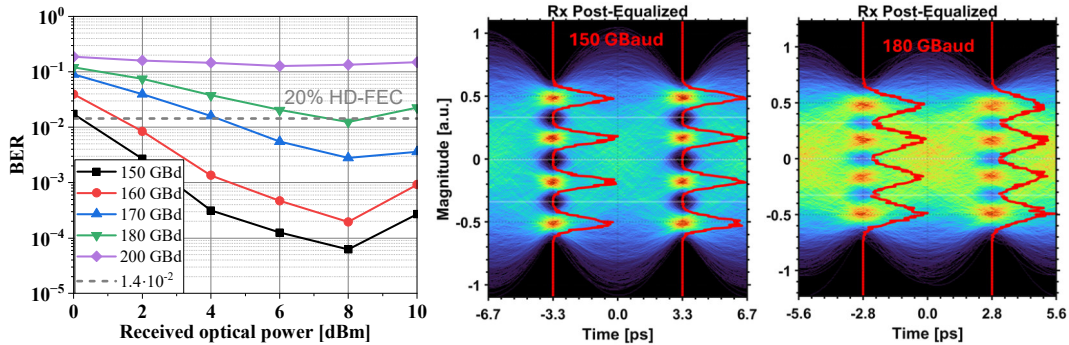


Fig. 3: BER measurement results of the PAM4 modulation scheme at different symbol rates (left). Eye diagram at 150 GBaud (center) and 180 GBaud (right).

4. Conclusion

A 76 GHz InP-based MZM with $V_{\pi} = 3.1$ V was co-packaged with a 224 Gbaud-class differential linear EML driver. The modulator front-end achieved 180 GBaud PAM4 transmission with a BER below the 20%-HD-FEC threshold with a low electrical energy per bit of 1.45 pJ/bit. The co-packaged MZM engine showed competitive performance [1-5] in terms of high symbol rate for IM/DD transmission links in the O-Band.

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